Fine-Grained Visual-Language Alignment for Remote Sensing Image–Text Retrieval

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***Abstract*—Remote sensing image–text retrieval (RSITR) is critical for applications, including environmental monitoring and disaster management. The main challenge in this field is that the multiscale feature of remote sensing images and the semantic differences of professional texts make it difficult to achieve accurate alignment. Existing coarse-grained methods struggle to address the inherent difference between images and text. In light of this, we propose the fine-grained visual-language alignment (FGVLA) method. Our FGVLA employs a hybrid loss function that combines coarse-grained contrastive and triplet loss with novel fine-grained loss. Fine-grained loss includes spatial mask loss and fine-grained contrastive loss to enhance semantic alignment. The method also introduces an inference process that works cooperatively with fine-grained loss to explicitly align image patches with textual nouns. Extensive experiments on RSICD, RSITMD, and UCM-Caption datasets demonstrate that FGVLA outperforms the existing methods, achieving superior retrieval performance. The code of our FGVLA has been released at https://github.com/Ji-Haoyang/FGVLA**

***Index Terms*—Fine-grained visual-language alignment (FGVLA), remote sensing image–text retrieval (RSITR), visual-language model.**

I. Introduction

REMOTE sensing image–text retrieval (RSITR) has become an essential task in fields, such as environmental monitoring [1], urban planning [2], and disaster management [3]. RSITR aligns the visual features of remote sensing images with the semantic representations of textual descriptions [4], [5]. Many factors, such as different object types and cluttered backgrounds, affect alignment accuracy. Obtaining a higher retrieval accuracy remains a critical challenge in RSITR [6].

Previous RSITR methods mainly focused on coarse-grained feature aligning, where both images and texts are represented by global embeddings. These methods use standard similarity metrics, such as cosine similarity or Euclidean distance, to compute the match between image and text features [7], [8]. Although these methods demonstrate efficacy in simple scenarios, their application to complex remote sensing datasets [4], [9] often results in compromised accuracy. These coarse-grained methods typically generate a global representation of images and texts, failing to capture detailed and localized features, which limits their capacity to account for fine-grained semantic relationships [10], [11]. Their inability to focus on specific regions or objects within an image that correspond to distinct textual elements renders these methods suboptimal for high-precision retrieval tasks.

Recent advancements in fine-grained RSITR have incorporated various methods, including multiscale feature fusion [12], attention mechanisms [13], [14], and transformer-based architectures [15], [16]. Global–local alignment techniques [7], [17] and contrastive learning strategies [18], [19] have significantly improved the semantic comprehension of remote sensing imagery, while knowledge-enhanced models [10], [20] attempt to bridge the domain-specific terminology gap. Spatial-channel attention mechanisms [14], [21] and prototype-based approaches [22], [23] have been introduced to strengthen visual-text alignment through discriminative learning. Despite these advancements, current fine-grained RSITR methods [24] face several challenges. Most approaches prioritize global feature alignment [12], [17] or partial region-text interactions [7], [13], leading to inadequate modeling of fine-grained relationships between localized image structures and textual descriptions. The absence of precise semantic grounding at the patch level hinders accurate retrieval in complex scenes with multiple salient objects [22], [25]. Moreover, the existing methods struggle with scale variations and exhibit limited robustness when faced with diverse object sizes and spatial distributions [19]. While adaptive margin strategies [26] and hierarchical aggregators [16] provide partial solutions, the intrinsic modality difference between pixel-

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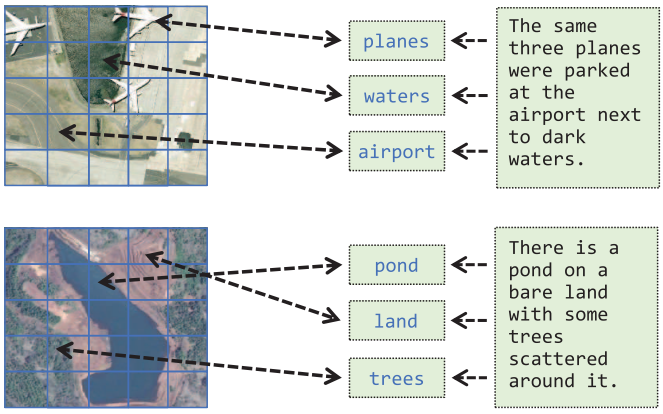


Fig. 1. Our motivation. We use a novel fine-grained loss to align text nouns and image patches, thereby enhancing the precision of image–text retrieval through improved local semantic correspondence.

level remote sensing representations and linguistic descriptors remains insufficiently addressed [7], [20]. These challenges highlight the critical need for innovative methods that effectively integrate both global and local features while ensuring the implementation of efficient fine-grained alignment mechanisms.

To address these challenges in traditional methods, we propose a fine-grained visual language alignment (FGVLA) method. The core idea of FGVLA is to use fine-grained semantic alignment to enhance the alignment between image patches and textual nouns, as shown in Fig. 1. Our method aims to address the challenge of image–text retrieval by combining global and local features of image–text pairs. First, FGVLA divides each image into smaller regions, and each aligns with its corresponding textual noun in the query. Then, our method adopts a hybrid loss function approach, which combines traditional coarse-grained loss functions, such as contrastive loss and triplet loss with newly introduced fine-grained loss functions. Notably, we introduce a fine-grained spatial mask loss and a fine-grained contrastive loss. The spatial mask loss guides the model to identify foreground features autonomously, suppress background noise, and provide accurate semantic focus. Fine-grained contrastive loss reduces the distance between relevant image patches and nouns in the feature space while suppressing irrelevant regions to enhance local feature discrimination. Therefore, the spatial mask loss provides high-quality positive and negative samples for contrastive loss, thereby optimizing the reliability of feature alignment. In turn, the contrastive loss further improves the accuracy of masked region selection by refining the feature distribution, jointly achieving multilevel fine-grained alignment. Finally, we also apply patch-noun alignment during inference to cooperate with the fine-grained loss during training.

In summary, our main contributions are as follows.

1) We propose an FGVLA method, which improves retrieval precision through patch-noun alignment, aligning image patches with textual nouns to ensure precise image–text retrieval.
2) We integrate traditional coarse-grained loss with novel fine-grained loss, including a fine-grained spatial mask loss and a contrastive loss, to enhance image–text alignment. This loss effectively captures both global semantic consistency and local feature relationships.
3) The inference process operationalizes this integration through a dual-path similarity computation: global embeddings establish coarse-grained correspondence, while local interactions between image patches and text nouns enable fine-grained alignment.
4) Our method has been extensively evaluated on three public datasets, RSICD, RSITMD, and and UCM-Caption. The experimental results indicate that our method achieves the state-of-the-art performance.

Our work is organized as follows. In Section II, we review the related work on RSITR and CLIP-based methods. In Section III, we provide a detailed description of our proposed method. In Section IV, we present extensive experimental results to demonstrate the effectiveness of our proposed approach. In Section V, we conclude the work and discuss potential directions for future research.

II. Related Work

In this section, the related works in two topics are mainly introduced: RSITR (see Section II-A) and CLIP-based RSITR methods (see Section II-B).

A. Remote Sensing Image–Text Retrieval

RSITR has undergone significant evolution, with methods progressing from coarse-grained to fine-grained retrieval techniques to achieve more accurate alignment between image and text modalities. Coarse-grained retrieval techniques [27] typically rely on global image features and monolithic text representations. Early works in this area, such as those by Shi and Zou [28], focus on generating language descriptions for remote sensing images using deep learning techniques, laying the groundwork for general image–text alignment tasks based on global features. Similarly, Lu et al. [9] propose recurrent neural networks for image captioning, which map global image features to corresponding text descriptions, thereby enabling coarse-grained retrieval.

Further advancements in fine-grained retrieval are made by Yuan et al. [4], who introduce a fine-grained multiscale method using asymmetric multimodal feature matching networks. Despite the higher level of abstraction, this method improves the feature matching between image regions and text. Yuan et al. [12] also employs contrastive learning and knowledge distillation to improve semantic localization in RSITR, refining the coarse-grained matching process. One significant advancement comes from Tang et al. [13], who propose the interacting-enhancing feature transformer. This method enhances the alignment of local features between images and text by employing transformer-based interactions between image patches and their corresponding text. Another notable contribution came from Zhang et al. [26], who applies curriculum learning combined with hypersphere-based feature alignment to improve the fine-grained matching of remote

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sensing images and textual queries. Additionally, He et al. [29] propose a visual global-salient-guided network that uses attention mechanisms to focus on important image regions and dynamically fuse relevant textual information. These approaches enable more precise semantic alignment between fine-grained image patches and corresponding text elements, thereby enhancing retrieval performance. These advancements [30], [31] in fine-grained methods underscore the growing need for detailed feature extraction and semantic alignment between image regions and text segments.

Despite these developments, fine-grained methods [32], [33] still face challenges in aligning specific image patches with textual descriptions in a semantically meaningful manner [34]. This limitation motivates the introduction of our proposed FGVLA approach. Our method aims to refine fine-grained retrieval by explicitly aligning patches with their corresponding nouns in textual descriptions. This method bridges the semantic gap between low-level visual features (i.e., image patches) and high-level language concepts (i.e., nouns), ensuring that both modalities are aligned at a more granular level. By leveraging patch-noun semantic alignment, our proposed method improves both retrieval accuracy and semantic relevance, thereby enhancing the overall retrieval process.

B. CLIP-Based RSITR Methods

CLIP-based methods have demonstrated impressive performance in image–text retrieval tasks by learning joint embeddings for images and text through contrastive learning [35]. These methods [36], [37] have been successfully adapted to remote sensing applications, where large-scale labeled datasets are often scarce. The ability of CLIP to perform zero-shot retrieval and to align images with textual descriptions makes it a valuable framework for RSITR tasks, particularly in scenarios involving diverse and high-resolution data.

Building on the capabilities of CLIP, researchers have adapted the method to enhance image–text alignment in remote sensing applications [38], [39]. For example, Al Rahhal et al. [40] fine-tune CLIP for specific land types in remote sensing images, achieving improved retrieval accuracy under diverse geographical conditions. Similarly, Mi et al. [20] integrate knowledge-aware learning into CLIP, enabling the model to handle ambiguous textual queries and enhance semantic understanding between text and image. CLIP has also been modified to incorporate fine-grained feature extraction [41], with methods such as those proposed by Chen et al. [42] introducing multilevel semantic interaction alignment to further improve the precision of image–text retrieval.

Moreover, CLIP [43] has been extended to address the challenges in fine-grained retrieval. Cai et al. [44] incorporate attention mechanisms into CLIP [45], enabling the model to focus on relevant image regions and their corresponding textual descriptions. This capability is particularly important for remote sensing images that contain numerous objects or exhibit diverse environmental conditions [46]. Zhou et al. [47] integrate multiscale feature extraction with CLIP, allowing the model to better handle variations in image resolution and enhance retrieval performance for high-resolution remote sensing images. These advancements [48] demonstrate CLIP's adaptability for fine-grained alignment in RSITR tasks.

However, even with these improvements, challenges persist in achieving precise alignment between image regions and their corresponding textual components. This limitation motivates the introduction of the proposed FGVLA approach, which aims to enhance CLIP by aligning image patches with specific nouns in text descriptions. FGVLA refines the semantic aligning process by focusing on local image features and their direct textual counterparts, offering a novel method for improving fine-grained retrieval in remote sensing applications. The method builds upon the strengths of CLIP and advances it by incorporating more detailed, region-specific alignments, which are essential for high-precision retrieval.

III. Method

In this section, we present a description of our FGVLA. First, we introduce our approach starting with the architecture in Section III-A. Then, we focus on the incorporation of fine-grained spatial mask loss and contrastive loss in Sections III-B and III-C. Finally, we outline the objective function in Section III-D and the inference process in Section III-E.

A. Overall

The retrieval performance of remote sensing images is often hindered by complex backgrounds and the lack of prominent semantic targets. To address this issue, we propose an integrated approach to enhance retrieval performance. As illustrated in Fig. 2, the overall pipeline combines coarse-grained global feature alignment with fine-grained local feature alignment, guided by a set of loss functions specifically designed to optimize semantic consistency. This alignment strategy not only ensures comprehensive semantic understanding but also preserves fine-grained local details critical for high-precision retrieval. The detailed methodology is described as follows.

1) Coarse-Grained Feature Extraction and Loss: Given an image x_i, we utilize the CLIP image encoder $\mathcal{E}_{\text{IE}}$ to extract the global feature $z_i^g \in \mathbb{R}^{1\times d}$, where $d = 512$ as specified by CLIP. Similarly, for a given text t_i, we employ the CLIP text encoder $\mathcal{E}_{\text{TE}}$ to obtain the corresponding global text feature $z_t^g \in \mathbb{R}^{1\times d}$ for coarse-grained retrieval, as described in the following:

$$z_i^g = \mathcal{E}_{\text{IE}}(x), \quad z_t^g = \mathcal{E}_{\text{TE}}(t) \tag{1}$$

where z_i^g and z_t^g represent the global features of the image and text, respectively. To align these global features, we optimize the model parameters using a coarse-grained loss. The coarse-grained loss $\mathcal{L}_{\text{cg}}$ consists of a coarse-grained triplet loss $\mathcal{L}_{\text{cgt}}$ and a coarse-grained contrastive loss $\mathcal{L}_{\text{cgc}}$

$$\mathcal{L}_{\text{cg}} = \mathcal{L}_{\text{cgt}} + \mathcal{L}_{\text{cgc}}. \tag{2}$$

2) Fine-Grained Feature Extraction and Loss: For fine-grained retrieval, we apply the CLIP image encoder to extract 196 patch-level features, denoted as $z_i^l \in \mathbb{R}^{196\times d}$, where $d = 512$ and the input patch number is specified as 14×14 in the CLIP model. Similarly, we utilize the CLIP text encoder to extract 77 token-level features from the input text, resulting in $z_t^l \in \mathbb{R}^{77\times d}$, as CLIP specifies a maximum text length of

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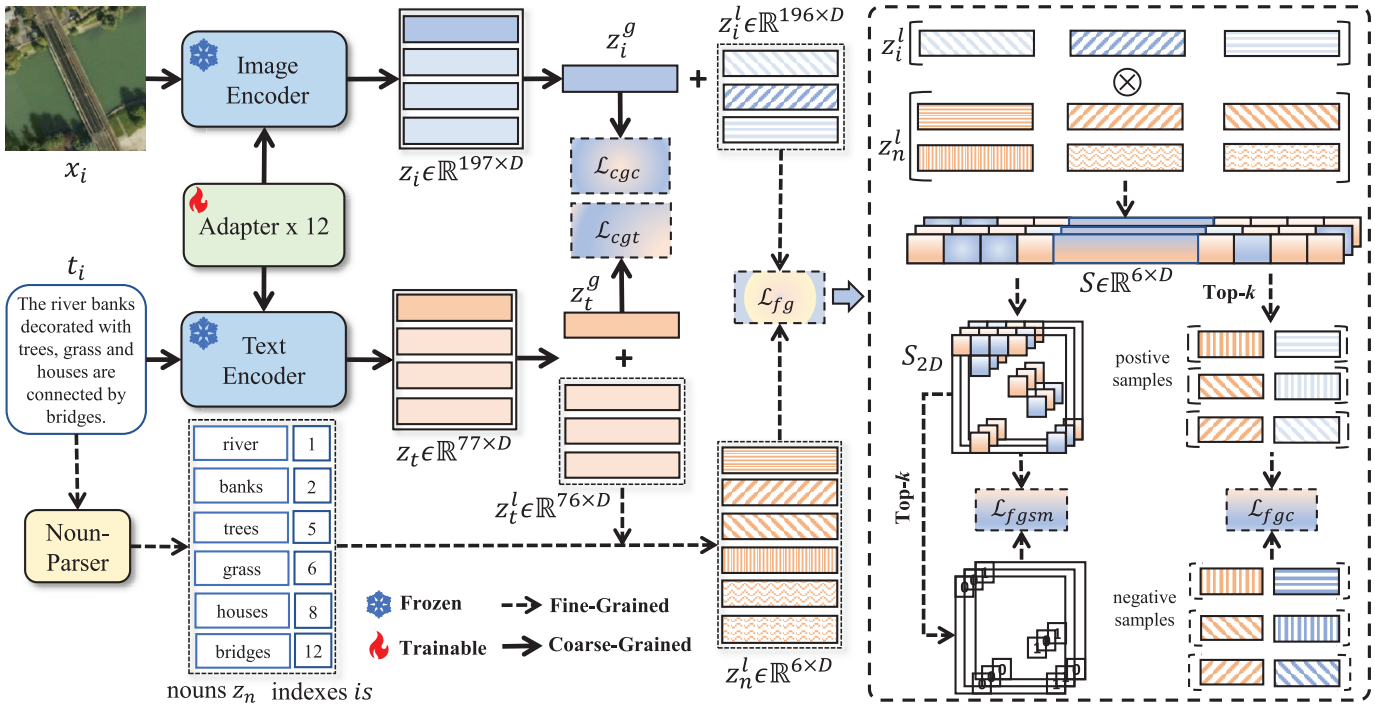


Fig. 2. Overall. (Left half) Illustrates the process of coarse-grained and fine-grained feature extraction. The ViT module and the BERT module process the image and text to obtain both global and local features. Nouns and their corresponding indexes are extracted from the original text using a noun parser, and these indexes are used to retrieve noun features from the local text features. The global features contribute to the computation of the coarse-grained contrastive loss and triplet loss, while the local image features and noun features are used to calculate the fine-grained loss. The right half of the framework details the computation process of the fine-grained loss. Specifically, the similarity matrix between image local features and text noun features is first computed, and then, the fine-grained spatial mask loss and contrastive loss are calculated based on the similarity values, following a Top-k selection principle.

77 tokens. To focus on semantically meaningful regions, we employ a noun-parser NP to extract all nouns z_n and record their indexes is. Using these indexes is, we retrieve a subset of token features corresponding to the nouns, represented as $z_n^l \in \mathbb{R}^{n_{\text{nou}}\times d}$, where n_{nou} denotes the number of nouns in the given text. This process is formalized as

$$\begin{aligned} z_i^l &= \mathcal{E}_{\text{IE}}(x), \quad z_t^l = \mathcal{E}_{\text{TE}}(t) \\ z_n, \text{is} &= \text{NP}(t), \quad z_n^l = z_t^l[\text{is}] \end{aligned} \tag{3}$$

where z_i^l and z_t^l denote the patch-level image features and token-level text features, respectively, NP refers to the NLTK toolkit [49], which is used to extract noun tokens from the input text, and is represents the indexes of these nouns. To align image patches with corresponding textual noun features, we optimize the model parameters using a fine-grained loss. The fine-grained loss $\mathcal{L}_{\text{fg}}$ consists of a fine-grained spatial mask loss $\mathcal{L}_{\text{fgsm}}$ and a fine-grained contrastive loss $\mathcal{L}_{\text{fgc}}$

$$\mathcal{L}_{\text{fg}} = \lambda \mathcal{L}_{\text{fgc}} + (1-\lambda)\mathcal{L}_{\text{fgsm}}. \tag{4}$$

3) Efficient Adapter: To enable efficient transfer from the natural image domain to the remote sensing image domain, we adopt an efficient adapter architecture, as illustrated in Fig. 3. Both the ViT and BERT backbones consist of 12 transformer blocks. Within each transformer block, we integrate a trainable adapter module to facilitate domain adaptation. We take the

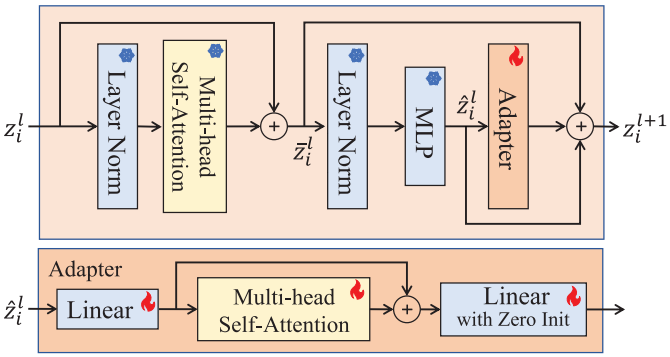


Fig. 3. Efficient adapter. (Top) Illustrates a standard transformer block augmented with a trainable adapter, while keeping the rest of the parameters frozen. (Bottom) Depicts the structure of the adapter. Specifically, the input first passes through an up-projection linear layer, followed by a residual connection to a multihead self-attention module. Finally, the output is passed through a downprojection linear layer, whose weights are initialized to zero.

ViT architecture as an example and define the forward process of each block as follows:

$$\begin{cases} \bar{z}_i^l = z_i^l + \mathcal{E}_{\text{MSA}}\left(\mathcal{E}_{\text{LN}}\left(z_i^l\right)\right) \\ \hat{z}_i^l = \mathcal{E}_{\text{MLP}}\left(\mathcal{E}_{\text{LN}}\left(\bar{z}_i^l\right)\right) \\ z_i^{l+1} = \bar{z}_i^l + \hat{z}_i^l + \mathcal{E}_{\text{Adapter}}\left(\hat{z}_i^l\right) \end{cases} \tag{5}$$

where $\mathcal{E}_{\text{LN}}$ denotes the layer normalization layer, $\mathcal{E}_{\text{MSA}}$ represents the multihead self-attention mechanism, $\mathcal{E}_{\text{MLP}}$ refers to the feedforward network, and $\mathcal{E}_{\text{Adapter}}$ indicates the lightweight trainable adapter module.

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B. Fine-Granined Spatial Mask Loss

Existing retrieval methods often struggle to distinguish salient image regions from irrelevant background content, leading to misalignment with corresponding textual descriptions. To address this limitation, we propose a fine-grained spatial mask loss that suppresses redundant background information while emphasizing key semantic regions. This loss enhances image–text alignment by guiding the model to focus on discriminative features that are critical for accurate retrieval.

First, we compute the similarity matrix $S \in \mathbb{R}^{n_{\text{nou}} \times 196}$ by measuring the similarity between the noun features z_n^l and the image patch features z_i^l

$$S = z_n^l \cdot z_i^{l\mathrm{T}}. \tag{6}$$

Then, the similarity matrix S is reshaped into a 2-D format $S_{\text{2-D}} \in \mathbb{R}^{n_{\text{nou}} \times 14 \times 14}$. To obtain a unified similarity map, we perform aggregation operations (e.g., mean pooling or max pooling) along the first dimension (i.e., the noun axis), resulting in an aggregated similarity map $S_{\text{2-D}}^{\text{agg}} \in \mathbb{R}^{1 \times 14 \times 14}$.

Next, we construct a binary mask $\mathcal{M} \in \mathbb{R}^{1 \times 14 \times 14}$ by selecting the Top-k highest values from the aggregated similarity map $S_{\text{2-D}}^{\text{agg}}$. Specifically, the mask assigns a value of 1 to the Top-k most similar entries and 0 to all remaining entries.

Finally, we compute the spatial mask loss using binary cross-entropy (BCE) between the aggregated similarity map $S_{\text{2-D}}^{\text{agg}}$ and the mask $\mathcal{M}$. The loss is defined as

$$\mathcal{L}_{\text{fgsm}} = \text{BCE}\left(S_{\text{2-D}}^{\text{agg}}, \mathcal{M}\right). \tag{7}$$

C. Fine-Granined Contrastive Loss

Traditional contrastive learning methods primarily optimize global features, often failing to capture fine-grained semantic correspondences. In image–text retrieval task, the specific image regions typically align with particular textual entities; however, misalignment between them can lead to retrieval errors. Moreover, the presence of irrelevant background information further hinders effective feature alignment. To address these issues, we propose a fine-grained contrastive loss that explicitly promotes accurate alignment between image patches and their corresponding textual nouns.

First, we continue with the similarity matrix $S \in \mathbb{R}^{n_{\text{nou}} \times 196}$ defined above. For each noun feature, we compute a contrastive loss. Specifically, the Top-k most similar image patches are selected as positive samples, while the remaining patches serve as potential negative samples. Then, we calculate the similarity between the anchor (i.e., the noun feature) and both positive and negative samples. The positive similarity scores l_{pos} are obtained by computing the dot product between the noun feature z_n^l and the corresponding Top-k positive patch features. Similarly, the negative similarity scores l_{neg} are computed between the noun feature and all patch features.

For each noun index i, the similarity scores between the ith noun feature and all image patch features are extracted from the similarity matrix S as

$$s_i = S\,[i, :] \tag{8}$$

where $s_i \in \mathbb{R}^{N_1}$ represents the similarity scores between the ith noun and all 196 image patches, and N_1 represents the number of patches of the image(196). The Top-k most similar patches are selected as positive samples

$$\{p_1, p_2, \ldots, p_k\} = \text{Top}K\,(s_i, k) \tag{9}$$

where $p_j \in \mathbb{R}^d$. The remaining patches are considered negative samples. Let $\mathbf{P}_{\text{pos}} \in \mathbb{R}^{k \times d}$ be the matrix formed by stacking the Top-k positive patch features $\{p_1, \ldots, p_k\}$, and let $\mathbf{P}_{\text{neg}} \in \mathbb{R}^{(196-k) \times d}$ be the matrix of negative patch features. The similarity between the anchor noun and the positive or negative patch groups is computed as

$$l_{\text{pos}} = \mathbf{P}_{\text{pos}} \cdot \left(\mathbf{z}_n^l\,[i]\right)^T, \quad l_{\text{neg}} = \mathbf{P}_{\text{neg}} \cdot \left(\mathbf{z}_n^l\,[i]\right)^T \tag{10}$$

where i represents the index subscript; then, we construct fine-grained contrastive loss [50] by connecting positive and negative similarities. The logits for the fine-grained contrastive loss are concatenated

$$l_{\text{gi}} = \left[l_{\text{pos}}, l_{\text{neg}}\right] \tag{11}$$

where $l_{\text{gi}} \in \mathbb{R}^{196}$; the labels for the positive samples are set to 1 and for negative samples are set to 0

$$\hat{l}\left[j\right] = \begin{cases} 1, & j \in \{1, 2, \ldots, k\} \\ 0, & j \in \{k+1, \ldots, 196\}. \end{cases} \tag{12}$$

Finally, we compute the fine-grained contrastive loss using the log-softmax of the logits. We sum the negative log-likelihood between the logits and labels to obtain the fine-grained contrastive loss, which encourages the method to maximize the similarity between correct patch–noun pairs while minimizing the similarity between incorrect pairs. The fine-grained contrastive loss is computed by

$$\mathcal{L}_{\text{fgc}} = -\sum_{j=1}^{N_1} \log\left(\frac{\exp\left(l_{\text{gi}}\left[j\right]\right)}{\sum_{i=1}^{N_1} \exp\left(l_{\text{gi}}\,[i]\right)}\right) \cdot \hat{l}\left[j\right] \tag{13}$$

where N_1 represents the number of patches of the image.

D. Object Function

The coarse-grained contrastive loss $\mathcal{L}_{\text{cgc}}$ and triplet loss $\mathcal{L}_{\text{cgt}}$ are designed to align global image and text representations. Combined with the fine-grained loss introduced above, we formulate the final training objective as follows.

The coarse-grained contrastive loss computes the similarity between image and text features using an inner product, scaled by a learnable temperature parameter. It enforces correct image–text pairs to exhibit the highest similarity by applying bidirectional cross-entropy. Specifically, for each image, the model predicts the corresponding text among all candidates and vice versa for each text. The coarse-grained contrastive loss $\mathcal{L}_{\text{cgc}}$ encourages correct pairs to receive high probabilities while suppressing mismatched pairs

$$\mathcal{L}_{\text{cgc}} = -\frac{1}{N_2}\sum_{i=1}^{N_2}\left(\log\frac{\exp\left(s_{ii}/\tau\right)}{\sum_{j=1}^{N_2}\exp\left(s_{ij}/\tau\right)} + \log\frac{\exp\left(s_{ii}/\tau\right)}{\sum_{j=1}^{N_2}\exp\left(s_{ji}/\tau\right)}\right) \tag{14}$$

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where N_2 is the size of the training set, τ is a learnable temperature parameter that controls the sharpness of the distribution, s_{ij} denotes the similarity between the ith image feature and the jth text feature, while s_{ii} corresponds to the similarity of a correctly matched image–text pair (i.e., when $i = j$).

The coarse-grained triplet loss ensures that the similarity of a correctly matched image–text pair exceeds that of mismatched pairs by at least a predefined margin m. It compares each positive pair against all negative samples, enforcing the constraint through a hinge function that filters out nonviolating cases and emphasizes hard negatives. The coarse-grained triplet loss is defined as

$$\begin{aligned}\mathcal{L}_{\text{cgt}} &= \frac{1}{2}\left(\mathcal{L}^1_{\text{cgt}} + \mathcal{L}^2_{\text{cgt}}\right)\\ \mathcal{L}^1_{\text{cgt}} &= \sum_{i\neq j} \max\left(0,\, m + s_{ij} - s_{ii}\right)\\ \mathcal{L}^2_{\text{cgt}} &= \sum_{i\neq j} \max\left(0,\, m + s_{ji} - s_{ii}\right)\end{aligned} \tag{15}$$

where $\mathcal{L}^1_{\text{cgt}}$ penalizes cases where the similarity s_{ij} between mismatched image–text pairs exceeds the similarity s_{ii} of the correct pair by more than the margin m. Similarly, $\mathcal{L}^2_{\text{cgt}}$ applies the same constraint in the reverse direction.

We also introduce an innovative fine-grained loss function, consisting of fine-grained spatial mask loss and fine-grained contrastive loss, which particularly focuses on fine-grained features extracted from image patches and textual nouns. The final loss $\mathcal{L}_{\text{total}}$, incorporating the coarse-grained triplet loss, the coarse-grained contrastive loss, and the fine-grained loss, is formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cgt}} + \mathcal{L}_{\text{cgc}} + \mathcal{L}_{\text{fg}}. \tag{16}$$

The final loss $\mathcal{L}_{\text{total}}$ integrates both coarse-grained and fine-grained information, thereby enhancing the robustness of the image–text retrieval model and improving performance in both global and detailed aligning.

E. Inference

In this section, we introduce the inference procedure for computing similarity scores between images and texts. The process integrates both coarse-grained and fine-grained similarity computations: global features are used to calculate coarse-grained similarity, while patch-level image features and textual noun features form the basis for fine-grained similarity. We aggregate fine-grained similarities using various pooling strategies, and the final similarity score is obtained by combining both coarse-grained and fine-grained components.

First, the coarse-grained similarity between the image and text is computed using the global features. The similarity matrix for the global features is calculated as the dot product of the image and text global features

$$s^g = z^g_i \cdot z^g_t \tag{17}$$

where $s^g \in \mathbb{R}^{N_i \times N_t}$ (N_i and N_t represent the number of images and texts, respectively) captures the overall similarity between each image and text.

Next, we calculate the fine-grained similarity of the image and text in both directions (text-to-image and image-to-text). In each direction, the similarity between patches and nouns is computed, followed by the aggregation of local similarities as

$$s^l_{t2i} = z^l_n \cdot z^{l\,\mathrm{T}}_i, \quad s^l_{i2t} = z^l_i \cdot z^{l\,\mathrm{T}}_n. \tag{18}$$

Then, we aggregate the fine-grained similarity matrix into fine-grained similarity results by mean or max.

Finally, we derive the final similarity scores by combining coarse-grained and fine-grained similarities. The final similarity matrix in both directions is computed as a weighted sum of the global and local similarities

$$\begin{aligned}s_{t2i} &= \alpha \cdot s^g + (1-\alpha)\cdot s^l_{t2i}\\ s_{i2t} &= \alpha \cdot s^g + (1-\alpha)\cdot s^l_{t2i}\end{aligned} \tag{19}$$

where α is a hyperparameter that balances the importance of coarse-grained and fine-grained similarities.

IV. Experiments

In this section, we first describe the dataset, evaluation metrics, and experimental details in Sections IV-A and IV-B. Next, we compare our method to related methods in Section IV-C. Then, we perform sufficient ablation and hyperparameter analysis in Section IV-D. After, we further analyze by visualization results in Section IV-E. Additionally, we conduct an efficiency analysis in Section IV-F. Finally, we analyze interesting examples about noun parsing quality and visualize attention about our method in Sections IV-G and IV-H.

A. Dataset and Evaluation Metrics

We evaluate our proposed FGVLA on the three widely used RSITR datasets: RSICD [9], RSITMD [4], and UCM-Caption [51]. The RSICD dataset contains 10921 image–text pairs, with each image having a resolution of 224 × 224. The RSITMD dataset, containing 4743 image–text pairs, has a higher resolution of 256 × 256 for each image. The UCM-Caption dataset is the earliest RS image–text datasets, which contains 2100 image–text pairs. Each image–text sample in these datasets consists of one image and five corresponding text description sentences. Consistent with the experimental details established by [52], we split the dataset into training, validation, and test sets in a ratio of 8:1:1.

We evaluate our method using two metrics: recall at K ($R@K$, where K = 1, 5, and 10) and mean recall (mR). $R@K$ measures the proportion of queries that successfully retrieve the ground truth among the top K results

$$R@K = \frac{1}{|Q|}\sum_{q_i \in Q} \mathbf{1}\left(L(q_i, K) \cap G(q_i) \neq \emptyset\right) \tag{20}$$

where $Q = \{q_1, q_2, \ldots, q_{N_3}\}$ is the set of N_3 test queries. Given a query q_i, $L(q_i, K)$ denotes the top K results retrieved by the model, and $G(q_i)$ is the set of ground truths relevant to this query. The function $\mathbf{1}(\cdot)$ is an indicator function that is 1 if

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$L(q_i, K) \cap G(q_i) \neq \emptyset$ and 0 otherwise. The mR is the average $R@K$ for both text retrieval and image retrieval tasks

$$\text{mR} = \left(\underbrace{R@1 + R@5 + R@10}_{\text{Text Retrieval}} + \underbrace{R@1 + R@5 + R@10}_{\text{Image Retrieval}} \right) /6. \tag{21}$$

The mR metric evaluates the overall retrieval performance.

B. Implementation Details

All experiments are conducted on a single NVIDIA RTX 4090 GPU. We tune the hyperparameters and set the contrastive loss weight (λ) in the fine-grained loss to 0.6 and the coarse-grained loss weight to 0.9 during inference. For CLIP, the final features are projected into a 512-D embedding space. We adopt the AdamW optimizer with a learning rate of 4×10^{-4} and a weight decay of 0.04, along with a linear learning rate decay schedule. The model is trained for ten epochs on both datasets using a batch size of 256. Following the protocol in [4], we employ a fivefold cross-validation strategy to report average performance.

C. Comparisons With the State-of-the-Art Methods

We compare our FGVLA method with two categories of baselines: traditional methods and CLIP-based methods. The experiments are conducted on the RSICD, RSITMD, and UCM-Caption datasets. The traditional baselines include LW-MCR [12], AMFMN [4], SWAN [53], MSITA [10], SCAT-PRG [14], IEFT [13], KAMCL [16], VGSGN [29], DOVE [54], MSA [22], and PIR-ITR [55]. The CLIP-based baselines include zero-shot CLIP [35], linear-probe CLIP [35], full-FT CLIP [35], SkyCLIP [56], adapter [57], CLIP-Adapter [58], cross-modal adapter [59], UniAdapter [60], SingleLanguage [40], PE-RSITR [61], and HarMA [52]. The brief descriptions of related methods are provided in the following.

Traditional Methods. The feature extractors used in traditional methods are typically based on deep networks with nontransformer architectures. The upper part of Table I summarizes the experimental results of these methods on RSICD and RSITMD. In the following, we analyze and discuss three recent state-of-the-art RSITR methods in detail as follows.

1) DOVE [54] introduces a regional-oriented attention and a lightweight text-assistant to enhance visual-textual alignment, guided by a global semantic constraint.
2) MSA [22] proposes a multiscale alignment transformer and a multiscale semantic loss to align image features and localized text across different scales.
3) PIR-ITR [55] utilizes prior knowledge with spatial–temporal progressive attention encoders and a clusterwise affiliation loss to capture long-range dependencies.

CLIP-based Methods. The CLIP-based methods are built upon transformer architectures. The lower part of Table I presents the experimental results of these methods. To further demonstrate the effectiveness of our method, we select three representative CLIP-based methods for comparative analysis.

1) UniAdapter [60] distributes adapters across modalities with partial weight sharing and unified knowledge transfer to reduce tunable parameters.
2) PE-RSITR [61] applies PETL and modality-specific adapters to improve efficiency, using contrastive learning for cross-modal alignment.
3) HarMA [52] employs hierarchical mini adapters and a novel loss to enhance alignment and reduce modality clustering with minimal updates.

1) Results on RSICD: We first evaluate our method on the large-scale RSITR dataset RSICD. The left half of Table I presents the retrieval performance of each method on this dataset. Compared with the best existing results, our method achieves an improvement of 0.57% in *R@1* and 0.75% in *R@10* for Text Retrieval. Notably, for the *R@10* metric, the improvement reaches 1.75%. For Image Retrieval, *R@10* increases by 0.69%. In addition, the mR metric improves by 0.50%. Despite the less significant gains on RSICD caused by lower quality noun features, our method consistently outperforms the existing approaches and remains highly competitive.

2) Results on RSITMD: The right half of Table I presents the performance of each method on RSITMD, where our proposed FGVLA method demonstrates significant improvements over the state-of-the-art approaches. For Text Retrieval, *R@1* and *R@5* increase by 0.52% and 3.62%, respectively, while for Image Retrieval, the improvements are 1.37% and 0.85%, respectively. Notably, our method achieves the most substantial gains on the *R@10* metric, with improvements of 3.77% for Text Retrieval and 1.71% for Image Retrieval. Moreover, the mR metric increases by 2.05%, indicating overall enhancement in retrieval performance. Compared to RSICD, the RSITMD dataset contains more fine-grained sentence descriptions. This allows our method to extract more accurate noun features that align more effectively with visual patches, thereby achieving superior performance on the more complex RSITMD dataset.

3) Results on UCM-Caption: Table II presents the performance of each method on UCM-Caption, where our FGVLA method demonstrates improvements over the second-best results. Specifically, for image retrieval, the *R@1*, *R@5*, and *R@10* metrics increase by 0.38%, 0.67%, and 0.24%, respectively, and the *R@1* metric of text retrieval is enhanced by 1.58%. Notably, for text retrieval, the method improves *R@5* by 5.71% and *R@10* by 2.72%, and the overall mR metric also exhibits a significant enhancement of 3.64%, indicating a strong comprehensive performance improvement. This demonstrates that our method possesses significant generalization ability, performing excellently across various datasets.

D. Ablation Studies

In this section, we conduct detailed ablation experiments and hyperparameter analysis on two datasets.

1) Main Analysis of Ablation Experiments: We conduct ablation experiments on the RSITMD dataset to evaluate the contributions of different modules in the proposed FGVLA method, as shown in Table III. FGVLA comprises three core components: the fine-grained contrastive loss $\mathcal{L}_{\text{fgc}}$, the spatial mask loss $\mathcal{L}_{\text{fgsm}}$, and the fine-grained inference process

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TABLE I

COMPARISONS OF RETRIEVAL PERFORMANCE ON RSICD AND RSITMD TEST SET, + MEANS INCREASED ACCURACY, – MEANS DECREASED ACCURACY, UNDERLINE DENOTES THE SECOND-BEST RESULTS, AND BOX INDICATES THE BEST RESULTS

	Methods	Time	RSICD dataset							RSITMD dataset						
			Text Retrieval			Image Retrieval			mR	Text Retrieval			Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
Traditional	LW-MCR-u	2022	4.39	13.35	20.29	4.30	18.85	32.34	15.59	9.73	26.77	37.61	9.25	34.07	54.03	28.58
	AMFMN-fusion	2022	5.39	15.08	23.40	4.90	18.28	31.44	16.42	10.63	24.78	41.81	11.51	34.69	54.87	29.72
	SWAN	2023	7.41	20.13	30.86	5.56	22.26	37.41	20.61	13.35	32.15	46.90	11.24	40.40	60.60	34.11
	MSITA	2024	8.67	22.71	33.91	6.13	21.98	35.39	21.47	15.22	34.20	47.65	12.15	39.92	57.72	34.48
	SCAT-PRG	2024	10.78	25.92	38.66	7.87	25.31	39.78	24.72	16.62	32.57	48.32	11.74	41.50	60.87	35.27
	IEFT	2023	8.11	27.45	43.25	6.26	24.00	39.98	24.84	15.48	37.61	51.40	11.10	37.99	58.75	35.38
	KAMCL	2023	11.99	27.17	38.33	7.78	25.23	40.02	25.08	16.81	34.96	47.12	14.60	41.86	59.86	35.87
	VGSGN	2024	8.33	21.87	32.57	6.53	23.13	36.85	21.55	14.16	34.96	50.66	13.23	42.57	63.41	36.50
	DOVE	2024	8.66	22.35	34.95	6.04	23.95	40.35	22.72	16.81	36.80	49.93	12.20	49.93	66.50	37.73
	MSA	2024	9.52	25.25	35.32	6.55	24.43	39.63	23.45	15.93	38.50	50.88	14.96	45.22	61.86	37.89
	PIR-ITR	2024	10.89	26.17	37.79	7.17	25.07	41.06	24.69	18.36	42.04	55.53	13.36	44.47	61.73	39.25
CLIP-based	Zero-shot CLIP	2021	6.77	15.37	23.15	5.01	15.75	24.21	15.04	9.29	26.33	37.39	7.79	23.67	38.89	23.89
	SkyCLIP	2024	6.59	16.10	26.53	7.14	22.34	34.29	18.83	10.18	25.44	35.62	10.88	33.27	49.82	27.54
	Adapter	2019	8.73	24.73	37.81	8.43	26.02	43.33	24.84	13.75	27.64	39.96	12.89	40.09	59.91	32.37
	CLIP-Adapter	2021	7.11	19.48	31.01	7.67	24.87	39.73	21.65	12.83	28.84	39.05	13.30	40.20	60.06	32.38
	Linear-probe CLIP	2021	8.46	24.41	37.72	7.81	25.89	42.47	24.46	17.02	33.12	48.35	13.33	41.80	63.89	36.25
	Cross-Modal Adapter	2022	11.18	27.31	40.62	9.57	30.74	48.36	27.96	18.16	36.08	48.72	16.31	44.33	64.75	38.06
	UniAdapter	2023	12.65	30.81	42.74	9.61	30.06	47.16	28.84	19.86	36.32	51.28	17.54	44.89	56.46	39.23
	SingleLanguage	2022	10.70	29.64	41.53	9.14	28.96	44.59	27.42	19.69	40.26	54.42	17.61	49.73	66.59	41.38
	PE-RSITR	2024	14.13	31.51	44.78	11.63	33.92	50.73	31.12	23.67	44.07	60.36	20.10	50.63	67.97	44.47
	Full-FT CLIP	2021	13.54	30.83	43.46	11.55	33.14	49.83	30.39	24.16	47.12	61.28	20.40	50.53	68.54	45.33
	HarMA	2024	16.36	34.48	47.74	12.92	37.17	53.07	33.62	25.81	48.37	60.61	19.92	53.27	71.21	46.53
	FGVLA (Ours)	2025	16.93	36.23	48.49	12.72	36.61	53.76	34.12	26.33	51.99	64.38	21.77	54.12	72.92	48.58
			+0.57	+1.75	+0.75	-0.20	-0.56	+0.69	+0.50	+0.52	+3.62	+3.77	+1.37	+0.85	+1.71	+2.05

TABLE II

COMPARISONS OF RETRIEVAL PERFORMANCE ON UCM-CAPTION TEST SET, + MEANS INCREASED ACCURACY, – MEANS DECREASED ACCURACY, UNDERLINE DENOTES THE SECOND-BEST RESULTS, AND BOX INDICATES THE BEST RESULTS

	Methods	Time	UCM-Caption dataset						
			Text Retrieval			Image Retrieval			mR
			R@1	R@5	R@10	R@1	R@5	R@10	
Traditional	LW-MCR-b	2022	12.38	43.81	59.52	12.00	46.38	72.48	41.10
	LW-MCR-d	2022	15.24	51.90	62.86	11.90	50.95	75.24	44.68
	LW-MCR-u	2022	18.10	47.14	63.81	13.14	50.38	79.52	45.35
	AMFMN-sim	2022	14.76	49.52	68.10	13.43	51.81	76.48	45.68
	AMFMN-soft	2022	12.86	51.90	66.67	14.19	51.71	78.48	45.97
	AMFMN-fusion	2022	16.67	45.71	68.57	12.86	53.24	79.43	46.08
	MSITA	2024	16.86	49.33	73.33	14.29	57.16	91.58	50.43
	MSA	2024	13.33	59.05	77.14	13.14	57.52	92.48	52.11
CLIP-based	Zero-shot CLIP	2021	10.95	34.76	55.71	7.33	35.14	54.57	33.08
	CLIP-Adapter	2021	9.83	42.52	66.79	14.61	51.03	84.14	44.82
	Linear-probe CLIP	2021	13.33	52.71	77.62	14.43	59.42	90.28	51.30
	Full-FT CLIP	2021	17.14	55.24	79.52	13.90	56.95	91.81	52.43
	Adapter	2019	10.17	49.73	80.08	18.20	62.47	95.77	52.74
	Cross-modal Adapter	2022	13.77	50.57	81.41	17.43	61.45	95.48	53.62
	SingleLanguage	2022	19.04	53.33	77.61	19.33	64.00	91.42	54.12
	UniAdapter	2023	14.46	55.19	83.95	16.74	61.43	95.76	54.59
	PE-RSITR	2024	22.71	55.81	80.33	18.82	62.84	93.72	55.71
	FGVLA (Ours)	2025	24.29	64.76	86.67	19.71	64.67	96.0	59.35
			+1.58	+5.71	+2.72	+0.38	+0.67	+0.24	+3.64

inference. We systematically assess their impact by performing eight ablation settings: starting from the baseline model, incrementally adding one module, combining any two modules, and finally integrating all three components. In all cases, the retrieval performance improves over the baseline. The best result is achieved when all three modules are enabled, with the mR metric reaching 48.58. These results demonstrate that the joint use of $\mathcal{L}_{\text{fgc}}$, $\mathcal{L}_{\text{fgsm}}$, and inference yields better performance than any individual component, highlighting the synergistic effectiveness of the proposed method.

2) Ablation Study of Top-k: We analyze the impact of the hyperparameter k used in the fine-grained loss, as shown in Tables IV and V. We first investigate the effect of varying the Top-k value in the spatial mask loss $\mathcal{L}_{\text{fgsm}}$, while keeping all other modules active. As shown in Table IV, the aggregation strategy is fixed to mean pooling, and k is varied from 4 to 7. The results show that the highest retrieval accuracy of 48.58 is achieved when $k = 5$, followed by 48.16 when $k = 4$. This indicates a unimodal performance trend, where retrieval performance peaks at $k = 5$ and declines when k deviates

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TABLE III

ABLATION STUDY ON DIFFERENT STRUCTURES OF FGVLA ON THE RSITMD TEST SET. THE ABSENCE OF ✓ IN THE INFERENCE COLUMN INDICATES THE USE OF THE STANDARD INFERENCE PROCESS

$\mathcal{L}_{fgc}$	$\mathcal{L}_{fgsm}$	$inference$	Text Retrieval			Image Retrieval			mR (%)
			R@1	R@5	R@10	R@1	R@5	R@10	
baseline			25.66	46.24	60.18	21.15	53.45	72.57	46.54
✓			26.33	51.33	62.83	22.3	53.94	72.79	48.25
	✓		27.21	49.78	62.61	22.12	53.67	72.65	48.01
		✓	26.77	50.22	63.27	21.77	53.89	73.23	48.19
✓	✓		26.55	49.12	64.82	21.55	54.25	72.70	48.16
✓		✓	26.99	50.44	63.72	22.39	54.12	72.61	48.38
	✓	✓	27.43	49.56	63.72	22.74	54.16	72.21	48.30
✓	✓	✓	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**

TABLE IV

EFFECT OF TOP-k IN SPATIAL MASK LOSS

k	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
4	25.66	51.77	63.05	21.37	53.94	73.19	48.16
5	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**
6	26.77	50.44	62.61	21.64	53.81	72.57	47.97
7	26.11	50.66	63.27	21.15	54.03	72.04	47.88

TABLE V

EFFECT OF TOP-k IN CONTRASTIVE LOSS

K	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
4	26.33	50.66	63.05	22.04	54.03	71.59	47.95
5	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**
6	26.99	50.66	63.50	22.08	54.07	72.48	48.30
7	26.77	50.88	63.50	21.06	54.69	72.30	48.20

TABLE VI

ABLATION EXPERIMENT OF HYPERPARAMETER α

α	1 - α	Text Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
0.9	0.1	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**
0.8	0.2	26.33	51.77	63.50	20.97	54.25	72.83	48.27
0.7	0.3	26.33	50.00	64.60	20.88	54.16	72.61	48.10
0.6	0.4	27.65	49.56	63.72	20.44	54.16	72.12	47.94

TABLE VII

ABLATION EXPERIMENT OF HYPERPARAMETER λ

λ	1 - λ	Text Retrieval			Image Retrieval			mR
		R@1	R@5	R@10	R@1	R@5	R@10	
0.9	0.1	26.55	50.44	64.16	22.52	53.58	72.35	48.27
0.8	0.2	26.33	50.88	62.61	20.93	53.81	73.05	47.94
0.7	0.3	26.11	50.44	62.17	21.59	53.81	72.35	47.74
0.6	0.4	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**
0.5	0.5	27.43	49.78	62.39	21.95	54.25	72.65	48.08

from this value. Next, we examine the effect of varying the Top-k value in the fine-grained contrastive loss $\mathcal{L}_{\text{fgc}}$, again with all modules enabled, as shown in Table V. By adjusting k from 4 to 7, we observe a similar trend: the highest retrieval performance of 48.58 occurs at $k = 5$, followed by 48.30 at $k = 6$. These results further confirm the unimodal behavior, with optimal performance achieved when $k = 5$.

3) Ablation Study of Hyperparameters α and λ: Tables VI and VII examine two key hyperparameters: the inference weight α (balancing coarse-grained and fine-grained similarity) and the weight λ within the fine-grained loss. In Table VI, we vary α while keeping all modules enabled and using mean aggregation. The highest mR (48.58) is achieved with a coarse-grained weight α of 0.9, indicating that global features are crucial for inference, while overreliance on fine-grained features leads to performance drop. Table VII explores the weight allocation between $\mathcal{L}_{\text{fgc}}$ and $\mathcal{L}_{\text{fgsm}}$, with all other settings fixed. The optimal result (mR =48.58) is obtained when their weights are set to 0.6 and 0.4, respectively, confirming the effectiveness of balanced fine-grained supervision.

TABLE VIII

ABLATION OF AGGREGATION OPERATIONS IN THE SPATIAL MASK LOSS

Aggregate	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
mean	26.33	51.99	63.38	21.77	54.12	72.92	**48.58**
max	27.21	50.22	62.17	21.50	53.85	72.39	47.89

TABLE IX

ABLATION OF AGGREGATION OPERATIONS IN INFERENCE

Aggregate	Text Retrieval			Image Retrieval			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
mean	26.33	51.99	64.38	21.77	54.12	72.92	**48.58**
max	26.11	51.11	62.61	21.33	54.34	73.14	48.10

4) Ablation Study of Aggregation Operations: We perform ablation studies on the impact of different aggregation functions, and the experimental results are in Tables VIII and IX. In both experiments, we retain the optimal settings from the previous studies and vary only the aggregation function. The results show that mean aggregation consistently outperforms max aggregation. This suggests that the mean function provides greater robustness to noise, while the max function is more susceptible to outliers and noisy activations.

E. Qualitative Analysis

In the following, we verify the effectiveness and training stability of our method in aligning visual and textual features and improving retrieval accuracy from aspects, such as visualization results, feature distribution, and loss curves.

1) Feature Visualization: To better interpret the learned feature space, we visualize the image and text embeddings using the t-SNE algorithm, as illustrated in Fig. 4. Blue and green represent image and text embeddings, respectively. The three subfigures correspond to training epochs 1, 5, and 10. At the initial training stage (epoch 1), the embeddings exhibit

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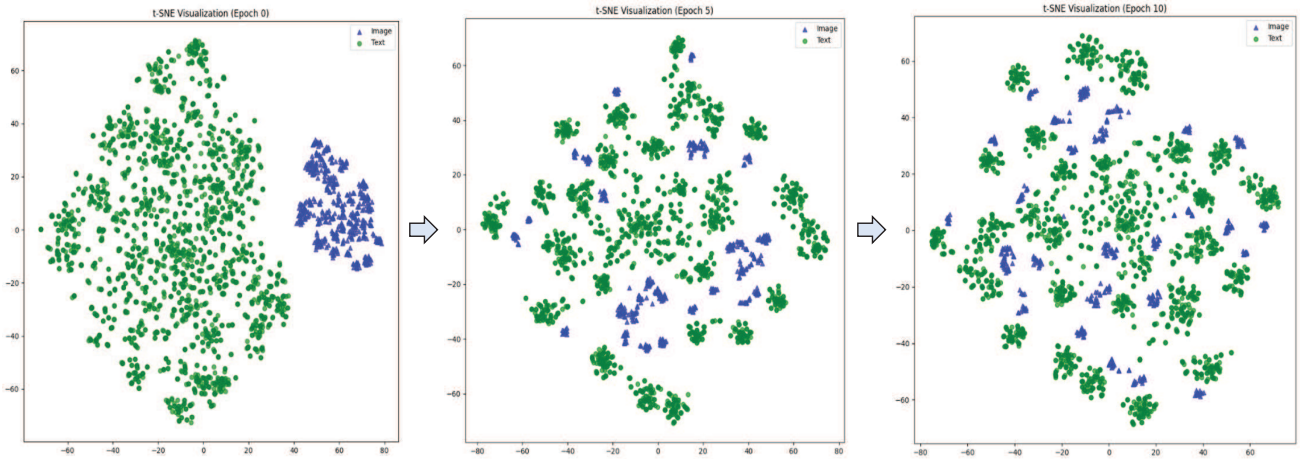


Fig. 4. Feature visualization. We employ t-SNE to project image and text features from the latent space onto a 2-D plane for visualization. Blue represents the image embeddings and green represents the text embeddings. From left to right, the three subfigures correspond to epochs 1, 5, and 10, respectively.

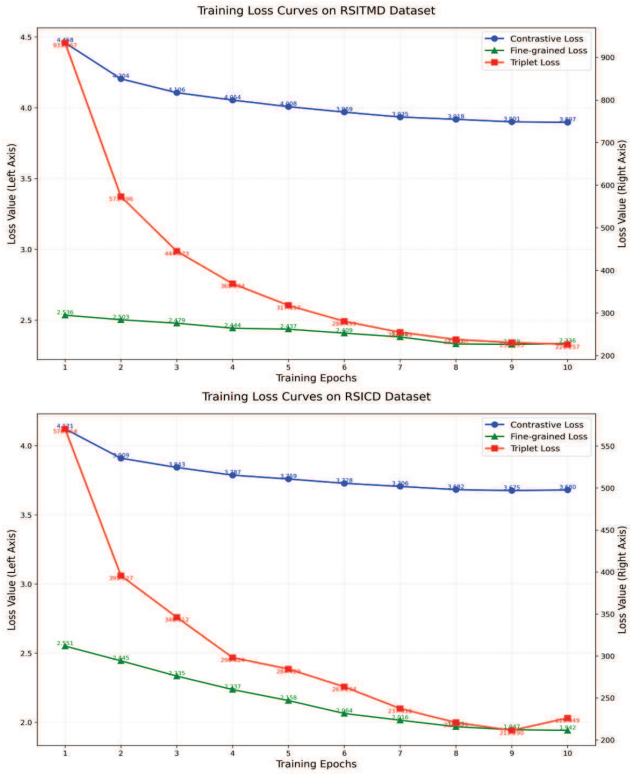


Fig. 5. Loss visualization. The training curves of the coarse-grained contrastive loss, triplet loss, and fine-grained loss on RSITMD and RSICD.

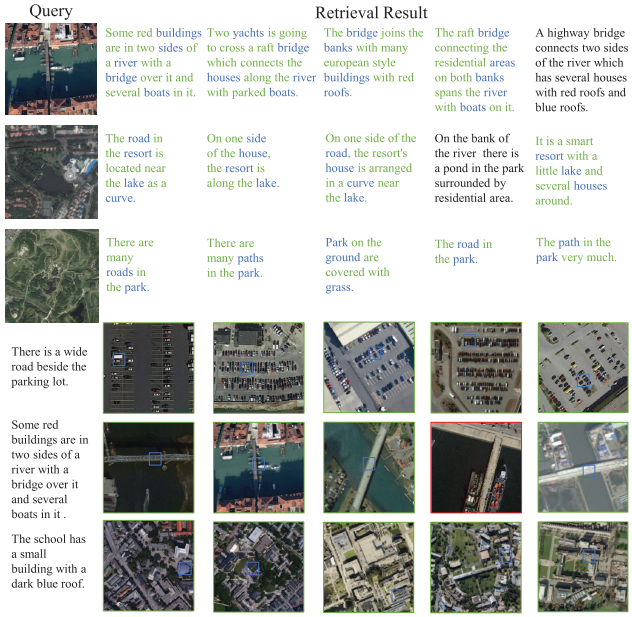


Fig. 6. Retrieval result visualization. (Top) Shows the image-to-text retrieval results on the RSITMD test set. We present the top-5 retrieved sentences. Green words indicate a correct match, blue words highlight the identified noun, and black words indicate an incorrect match. (Bottom) Illustrates the text-to-image retrieval results. We also display the top-5 retrieved images. Green boxes indicate correct matches, red boxes indicate incorrect ones, and small blue boxes highlight noun-related image regions.

a scattered distribution, indicating weak semantic alignment between image and text features. As training progresses (epoch 5), the embeddings begin to form more compact clusters. By epoch 10, semantically similar image–text pairs are positioned closer together in the latent space. This trend shows the effectiveness of our method in learning well-aligned visual–textual representations, thereby enhancing retrieval performance.

2) Training Loss Convergence: To assess the training stability and convergence behavior of our method, we analyze the evolution of the coarse-grained contrastive loss, triplet loss, and fine-grained loss across training epochs on the RSITMD and RSICD datasets, as shown in Fig. 5. The results show that, as training progresses, the model learns increasingly refined feature alignments, and the loss values gradually stabilize. All loss functions exhibit clear convergence trends, demonstrating the effectiveness and stability of the optimization process.

3) Image-to-Text Retrieval Results: We evaluate the image-to-text retrieval performance on the RSITMD test set by retrieving the top-5 most relevant text sentences for each query image, as shown in Fig. 6. Correctly retrieved sentences are marked in green, while identified nouns are highlighted in blue. Incorrectly retrieved sentences are displayed in black. The visualization demonstrates that our model effectively retrieves semantically relevant text. However, in the case of

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TABLE X

ABLATION STUDY ON THE COMPUTATIONAL COST OF FGVLA ON THE RSITMD DATASET DURING TRAINING. THE METRICS ARE REPORTED IN SECONDS (S), DETAILING TIME PER BATCH FOR INDIVIDUAL COMPUTATIONAL STEPS AND TOTAL TIME FOR FULL EPOCHS (67 BATCHES PER EPOCH)

$\mathcal{L}_{fgc}$	$\mathcal{L}_{fgsm}$	*Adapter*	Forward (s)	Total Loss (s)	Backward (s)	1 epoch (s)	10 epoch (s)
		✓	0.34	0.0031	0.21	38.43	384.31
✓		✓	0.34	0.39	0.80	102.93	1029.32
	✓	✓	0.34	0.05	0.31	47.75	477.51
✓	✓	✓	0.33	0.42	0.82	103.54	1035.42

TABLE XI

INFERENCE EFFICIENCY ANALYSIS ON RSITMD, RSICD, AND UCM-CAPTION DATASETS. THE TABLE COMPARES PERFORMANCE WITH AND WITHOUT THE FINE-TUNING ADAPTER FOR BOTH TEXT RETRIEVAL AND IMAGE RETRIEVAL TASKS

Dataset	**Task**	**Without Adapter**		**With Adapter**	
		Time (s)	FPS	Time (s)	FPS
RSITMD	Text Retrieval (2260 queries)	3.0596	738.66	3.1799	710.72
	Image Retrieval (452 queries)	2.9824	151.55	3.1104	145.32
RSICD	Text Retrieval (5465 queries)	6.3048	866.80	6.5409	835.51
	Image Retrieval (1093 queries)	6.0323	181.19	6.2881	173.82
UCM-Caption	Text Retrieval (1050 queries)	1.9249	545.49	1.9803	530.23
	Image Retrieval (210 queries)	1.8869	111.29	1.9479	107.81

more complex sentences, model tends to focus primarily on noun phrases and ignore other informative attributes.

4) Text-to-Image Retrieval Results: We further evaluate the text-to-image retrieval performance by using a textual query to retrieve the top-5 most relevant images from the RSITMD test set, as illustrated in Fig. 6. Correctly retrieved images are highlighted with green bounding boxes, while incorrect retrievals are marked in red. The small blue boxes highlight the image area related to the nouns in the correct search result. The results demonstrate that our method identifies visually and semantically relevant images based on textual input. High retrieval accuracy is achieved when the query contains explicit spatial or object-related attributes. However, misretrievals occur when the query is ambiguous, when visually similar images belong to different semantic categories, or when certain described features are absent in the dataset.

F. Efficiency Analysis

We first analyze the computational overhead of our proposed modules during training time, as detailed in Table X. We conduct all experiments on the RSITMD dataset, setting the input image resolution to 224×224 and training with 67 batches per epoch with a batch size of 256. All computations are performed on a single NVIDIA 4090 GPU. The baseline only includes the adapter, providing a comparative benchmark for computational overhead; it requires 0.34 s for the forward and 0.21 s for the backward per batch, culminating in a total training time of 384.31 s for ten epochs. The introduction of $\mathcal{L}_{\text{fgc}}$ introduces the most significant computational overhead. This is primarily because $\mathcal{L}_{\text{fgc}}$ involves the alignment between dense image patches and text nouns pairs, which significantly increases the computation time of the total loss from 0.0031 to 0.39 s and, more importantly, increases the backward time from 0.21 to 0.80 s. Consequently, the total training time for 10 epochs increases to 1029.32 s. In contrast, incorporating $\mathcal{L}_{\text{fgsm}}$ results in a more moderate increase in cost, with a total training duration of 477.51 s. When both components are integrated, the total training time is 1035.42 s. This analysis reveals that while our proposed modules are crucial for performance gains, the $\mathcal{L}_{\text{fgc}}$ module is the primary contributor to the increased training duration, highlighting a clear tradeoff between retrieval accuracy and computational efficiency during training.

We further evaluate the inference efficiency of our method across the RSITMD, RSICD, and UCM-Caption datasets, as detailed in Table XI. A performance disparity is observed between text retrieval and image retrieval. This efficiency gap stems from our fine-grained inference design. Text retrieval is fast because each image is represented by a fixed number of patches, a uniform structure that enables highly parallelized computation. Consequently, our method demonstrates high throughput for the text retrieval, achieving up to 710.72 queries per second (FPS) on RSITMD, 530.23 FPS on UCM-Caption, and 835.51 FPS on the larger RSICD dataset. In contrast, image retrieval, which matches an image against a text dataset, operates at a more moderate speed, reaching 107.81 FPS on UCM-Caption, 145.32 FPS on RSITMD, and 173.82 FPS on RSICD. We also note that the inclusion of our adapter introduces a minimal and acceptable computational overhead. For instance, on the RSITMD dataset, text retrieval is 738.66 FPS without the adapter and 710.72 FPS with it, indicating a negligible impact on efficiency. Overall, the results confirm that our method is highly efficient for practical deployment, especially for the text retrieval task.

G. Cases of Noun Parsing and Text Quality Analysis

We analyze the retrieval results of our method under various noun parsing qualities, as shown in Fig. 7. Our findings indicate that the method's retrieval result is indeed influenced by the quality of noun parsing. In most cases, when key nouns, such as "airplane" and "car," are parsed correctly, our method retrieves the correct results by effectively linking these nouns with the corresponding images. The method also exhibits a degree of robustness to minor parsing errors; for example, incorrectly identifying "roofed" as a noun when there are other

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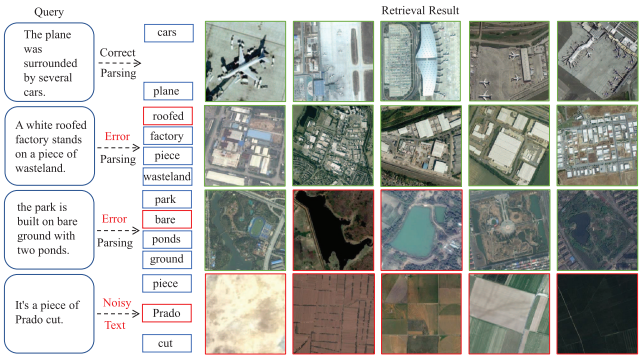


Fig. 7. Cases of noun parsing and text quality visualization. The figure illustrates four scenarios: correct noun parsing leads to successful retrieval; parsing errors have different impacts on retrieval accuracy; and noisy text containing irrelevant nouns leads to retrieval failure. Green borders indicate correctly retrieved images, while red borders indicate incorrect ones.

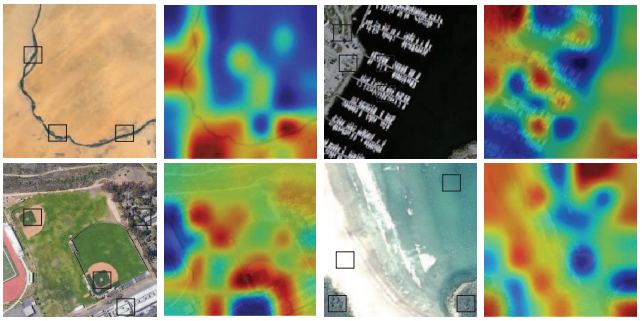


Fig. 8. Attention visualization. Visual comparison of the attention heatmap with manually annotated regions on the image. The black boxes in the original image represent the regions that humans pay attention to, the red in the heatmap represents the high attention regions of our method, and the blue represents the low attention regions.

related nouns, such as "factory," does not affect the retrieval results. However, more significant parsing errors, such as misidentifying the adjective "bare" as a noun, can mislead the fine-grained matching process and result in retrieval failures. Furthermore, the presence of semantic noise, exemplified by the out-of-context proper noun "Prado," leads to completely incorrect results even when parsing is technically correct.

H. Attention Analysis

We investigate how our proposed method attends to different image patches and compares its attentions with areas that humans typically focus on in complex scenes. We visualize the heatmap by calculating the similarity between each noun in the text and each patch in the image, as shown in Fig. 8. In these heatmaps, red indicates higher similarity, while blue indicates lower. To facilitate comparison with human attention, we manually drew black boxes on the original images to indicate the areas that humans usually focus on. We find significant correspondence between the attention of our method and the regions of interest annotated by humans. The regions with higher attention in our method usually overlap with or contain the regions marked by the black box. This consistency suggests that our method has successfully learned to prioritize visually salient regions and semantically meaningful objects in complex scenes, which are precisely the areas that humans primarily focus on.

V. Conclusion

In this work, we propose an FGVLA method for RSITR. Our method is optimized using a novel fine-grained loss function that combines spatial mask loss and contrastive loss through weighted integration. At inference time, we incorporate patch–noun semantic alignment to enhance retrieval precision. Extensive evaluations on three widely used datasets demonstrate the superiority of our method over other state-of-the-art methods. Furthermore, ablation studies and qualitative analyses confirm the effectiveness of our method in training, inference, and hyperparameter selection. Next, efficiency analysis shows that our method takes both accuracy and efficiency into consideration. Then, some examples show that the retrieval results are indeed affected by the quality of noun parsing and noisy text. Finally, attention analysis shows that in complex scenes, the regions our method focuses on are exactly the same as those where humans primarily focus. In future work, we aim to explore more advanced fine-grained alignment strategies to further improve retrieval performance in complex remote sensing scenarios.

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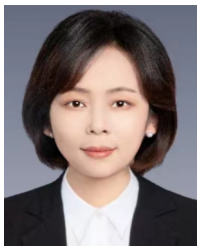
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